

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0092

Olympus 500 kV
April 24, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0092. The Interconnection Customer proposes a 1365.4 MW natural gas combined cycle facility interconnecting at the Olympus 500 kV substation in the Ohio area. The request was received on January 06, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 78 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$237,391,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of INTERNATIONAL PAPER - FATLICK 500kV, the GAVIN POWER LLC 345/230 kV XFMR loads to 184.7% of its 286 MVA rating. Project DFAX: 13.9%.

OLYMPUS 500/230 kV TX: 157.2% loading (789 MVA) following the outage of the FLATLICK - INTERNATIONAL PAPER 500kV; DFAX = 39.1%.

For the loss of INTERNATIONAL PAPER - FLATLICK 500kV, the SPORN - GAVIN POWER LLC 230kV loads to 145.9% of its 619 MVA rating. Project DFAX: 32.1%.

GAVIN POWER LLC 500-345 kV XFMR: 137.3% loading (829 MVA) following the outage of the INTERNATIONAL PAPER - FLATLICK 500kV; DFAX = 36.2%.

SHERWOOD FOREST 100/1 kV XFMR: 134.0% loading (356 MVA) following the outage of the INTERNATIONAL PAPER - FLATLICK 500kV Line; DFAX = 6.7%.

For the loss of FLATLICK - INTERNATIONAL PAPER 500kV L, the COMBEE RD - FLATLICK 230kV loads to 125.5% of its 532 MVA rating. Project DFAX: 16.3%.

The COMBEE RD - AXTELL SUBSTATION 230kV is overloaded to 125.5% of its 524 MVA emergency rating for the loss of the INTERNATIONAL PAPER - FLATLICK 500kV. The project contributes 16.3% DFAX (222.0 MW) to this overload.

For the loss of FLATLICK - INTERNATIONAL PAPER SUBSTATION 500kV L, the WHARTON - AXTELL 230kV loads to 124.7% of its 517 MVA rating. Project DFAX: 16.3%.

The OLYMPUS 230/1 kV Transformer is overloaded to 118.3% of its 242 MVA emergency rating for the loss of the FLATLICK - INTERNATIONAL PAPER 500kV. The project contributes 6.1% DFAX (83.0 MW) to this overload.

For the loss of INTERNATIONAL PAPER - FLATLICK 500kV L, the POMONA PARK SUBSTATION - OLYMPUS 138kV Line loads to 116.8% of its 246 MVA rating. Project DFAX: 6.1%.

FLATLICK 230-1 kV XFMR: 115.8% loading (988 MVA) following the outage of the INTERNATIONAL PAPER - FLATLICK SUBSTATION 500kV; DFAX = 23.0%.

The OLYMPUS 138/1kV Autotransformer is overloaded to 114.6% of its 250 MVA emergency rating for the loss of the FLATLICK - INTERNATIONAL PAPER 500kV. The project contributes 6.1% DFAX (83.0 MW) to this overload.

SHERWOOD FOREST 230-1 kV Transformer Bank: 114.1% loading (419 MVA) following the outage of the INTERNATIONAL PAPER - FLATLICK 500kV L; DFAX = 6.7%.

FLATLICK 500/1 kV Transformer: 107.7% loading (1,063 MVA) following the outage of the INTERNATIONAL PAPER - FLATLICK 500kV; DFAX = 23.0%.

STILES - OLYMPUS 230kV: 106.3% loading (476 MVA) following the outage of the INTERNATIONAL PAPER SUBSTATION - FLATLICK 500kV; DFAX = 12.0%.

STILES - 7TH AVENUE 230kV: 101.5% loading (484 MVA) following the outage of the FLATLICK - INTERNATIONAL PAPER 500kV Line; DFAX = 12.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GAVIN POWER LC 345/230 kV TX	Replace transformer with higher-rated unit	\$20,114,000	89% (\$17,880,000)
OLYMPUS 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$28,834,000	92% (\$26,428,000)
SPORN - GAVIN POWER LLC 230kV	Rebuild line with higher-capacity conductor	\$13,854,000	82% (\$11,365,000)
GAVIN POWER LLC 500-345 kV XFMR	Replace transformer with higher-rated unit	\$28,503,000	97% (\$27,530,000)
SHERWOOD FOREST 100 kV/1 kV TX	Replace transformer with higher-rated unit	\$2,926,000	100% (\$2,926,000)
COMBEE RD - FLATLICK 230kV	Rebuild line with higher-capacity conductor	\$42,608,000	78% (\$33,451,000)
COMBEE ROD - AXTELL 230kV	Rebuild line with higher-capacity conductor	\$36,104,000	78% (\$28,345,000)
AXTELL - WHARTON SUBSTATION 230kV line	Rebuild line with higher-capacity conductor	\$31,685,000	78% (\$24,876,000)
OLYMPUS 230/1 kV Transformer	Replace transformer with higher-rated unit	\$7,795,000	80% (\$6,242,000)
POMONA PARK SUBSTATION - OLYMPUS 138kV Line	Rebuild line with higher-capacity conductor	\$2,761,000	80% (\$2,211,000)
FLATLICK 230 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$8,484,000	83% (\$7,007,000)
OLYPUS 138/1 kV Transformer	Replace transformer with higher-rated unit	\$3,501,000	80% (\$2,803,000)
SHERWOOD FOREST 230/1 kV Autotransformer	Replace transformer with higher-rated unit	\$9,844,000	100% (\$9,844,000)
FLATLICK 500 kV/1 kV TX	Replace transformer with higher-rated unit	\$25,667,000	83% (\$21,197,000)
STILES - OLYMPUS 230kV L	Reconductor line	\$5,305,000	74% (\$3,897,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STILES - 7TH AVENUE 230kV	Reconductor line	\$15,504,000	74% (\$11,389,000)
	Total allocated to project		\$237,391,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.